

Niraj Umeshbhai Patel

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EDUCATION

Master of Science in Software Engineering Systems, GPA: 3.9

Northeastern University

Sep. 2024 – Jun. 2026

Boston, USA

Bachelor of Engineering in Electrical Engineering

Gujarat Technological University

Aug. 2018 – May 2022

Anand, India

TECHNICAL SKILLS

Languages: Java, JavaScript/Node.js, TypeScript, Python, SQL, Bash

Backend & APIs: Node.js, REST, GraphQL, FastAPI, PostgreSQL, MySQL

Distributed Systems & Messaging: Apache Kafka, Docker, Kubernetes, Zookeeper, containerd

Cloud & Infrastructure: AWS (EC2,S3,RDS,Lambda,IAM,KMS,VPC,ALB,EKS,CloudFormation), Azure (AKS,ADF), GCP, OCI

IaC & CI/CD: Terraform, Ansible, Packer, GitHub Actions, Azure DevOps, Harness, Jenkins, ArgoCD

Observability & Monitoring: Prometheus, Grafana, Datadog, CloudWatch

AI/ML Tooling: CrewAI, AutoGen, Llama 3.1 (llama.cpp), Hugging Face PEFT/QLoRA, Gemini 2.0 Flash, Hybrid RAG (FAISS+BM25), Claude Code, Microsoft Copilot

EXPERIENCE

Data Platform Engineer Intern

State Street

Jun. 2025 – Aug. 2025

Boston, USA

- Automated monthly credential rotation for **50+ Oracle Cloud** applications using **Python** and Ansible, saving 3+ hours monthly and eliminating rotation errors.
- Migrated **3 Terraform** codebases (25+ components) from Azure DevOps to **Harness** after pipeline runtime hit 90 minutes, cutting deploy time to 30 minutes.
- Provisioned an Azure jumpserver with **Terraform** and OCI integration to remove direct public exposure of data-platform resources, securing access for a 5-person team.

Assistant System Engineer

Tata Consultancy Services

Jul. 2022 – Aug. 2024

Gandhinagar, India

- Developed and maintained **RESTful and GraphQL APIs** using **Node.js**, integrating live third-party services into the LeanIX platform and reducing API integration time by 30% for downstream teams.
- Engineered backend automation workflows in **Java**, writing optimized **PostgreSQL** queries and automated email notification services, eliminating 5+ hours/week of manual curation tasks via Control-M scheduling.
- Deployed **end-to-end CI/CD pipelines** using **Azure DevOps** and GitHub Actions, reducing deployment lead time to under 3 hours and eliminating manual intervention across LeanIX workflows.
- Reduced container image sizes by **35%** by containerizing **10+ Java and Node.js** microservices with Docker multi-stage builds to shrink bloated single-stage images and speed rollout.
- Codified dev, staging, and production environments as reusable **Terraform IaC** modules to eliminate configuration drift, lowering environment setup time by **20%**.
- Configured **CloudWatch** and Azure Monitor alerting to meet availability targets during **24x7 on-call** rotation, reducing incident triage time by **30%** for critical LeanIX microservices.
- Designed batch and delta **ETL** pipelines on **Azure Data Factory** and the **Databricks Lakehouse** for a Fortune 500 MedTech client, running a **1TB** month-end full load plus daily delta refreshes across **40+ pipelines**.
- Mentored **3 interns** through structured onboarding and troubleshooting support, maintaining 2-week sprint cycle times and increasing team throughput by **28%**.

PROJECTS

Cloud-Native Web Application | *Node.js, PostgreSQL, AWS, Terraform, GitHub Actions*

- Built RESTful APIs in **Node.js** with PostgreSQL for a cloud-native application, implementing CRUD operations and structured error handling.
- Raised application availability ~30% by auto-scaling 3-5 **EC2** instances behind an Application Load Balancer on a CPU-based policy.

Multi-Cloud Cost Optimization Platform | *Python, FastAPI, Airflow, Docker, Kubernetes*

- Created a **FastAPI** service scanning **100+** AWS resources to detect idle capacity, flagging ~25% in potential monthly savings.
- Containerized the platform with **Docker** and Kubernetes, orchestrating daily scans in Airflow to enable horizontal scaling across 100+ resources.

CloudArch Designer – AWS Architecture Generator | *Python, Gemini 2.0 Flash, Terraform, FAISS/BM25*

- Implemented a **9-stage** NL-to-Terraform pipeline scoping **Gemini 2.0 Flash** to extraction/repair only, holding Terraform validity at 96%.
- Integrated a Stage 0 prompt-normalization step with hybrid RAG (FAISS + BM25), lifting consistency 0.446 to 0.914 across 180 test variants.

Distributed Event-Driven Registration System | *Python, Apache Kafka, Docker, Zookeeper*

- Enabled **20+** concurrent student registrations by integrating **Apache Kafka** pub/sub across 3 agents and 4 topics to decouple services.
- Designed a Zookeeper-coordinated deployment for fault-tolerant message persistence across the 4-topic, 3-agent **Kafka** pipeline.

Store Management Database System | *Oracle PL/SQL, SQL, ER Modeling*

- Established a **12-table** Oracle PL/SQL schema with **15 analytical views** covering inventory, sales, and customer data.
- Authored a CALCULATE_TOTAL_AMOUNT UDF with loyalty-discount logic and an automated reorder trigger, enforcing access across 4 user roles.

CERTIFICATIONS

- AWS Certified Developer – Associate
- AZ-104: Microsoft Azure Administrator